

Creation Date 23-May-2011

Revision Date 05-Jun-2024

Revision Number 8

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: **Phenol liquefied 80 w/w in water**  
Cat No. : **P/2315/PB17**

Unique Formula Identifier (UFI) **E9P8-R20T-VX0J-9U9G**

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

#### EU entity/business name

Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

#### UK entity/business name

Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

#### Swiss distributor - Fisher Scientific AG

Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

E-mail address **begel.sdsdesk@thermofisher.com**

### 1.4. Emergency telephone number

Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

For customers in Switzerland:

Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

#### Poison Centre - Emergency information services

**Ireland** : National Poisons Information Centre (NPIC) -  
**01 809 2166** (8am-10pm, 7 days a week)  
**Malta** : +356 2395 2000  
**Cyprus** : +357 2240 5611

## SECTION 2: HAZARDS IDENTIFICATION

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## 2.1. Classification of the substance or mixture

### CLP Classification - Regulation (EC) No 1272/2008

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

|                                                      |                     |
|------------------------------------------------------|---------------------|
| Acute oral toxicity                                  | Category 3 (H301)   |
| Acute dermal toxicity                                | Category 3 (H311)   |
| Acute Inhalation Toxicity - Vapors                   | Category 3 (H331)   |
| Skin Corrosion/Irritation                            | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation                    | Category 1 (H318)   |
| Germ Cell Mutagenicity                               | Category 2 (H341)   |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373)   |

#### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### **Hazard Statements**

- H301 + H311 + H331 - Toxic if swallowed, in contact with skin or if inhaled
- H314 - Causes severe skin burns and eye damage
- H341 - Suspected of causing genetic defects
- H373 - May cause damage to organs through prolonged or repeated exposure

### **Precautionary Statements**

- P280 - Wear protective gloves/protective clothing/eye protection/face protection
- P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
- P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
- P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
- P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
- P310 - Immediately call a POISON CENTER or doctor/physician

## 2.3. Other hazards

Toxicity to Soil Dwelling Organisms

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Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component | CAS No    | EC No             | Weight %  | CLP Classification - Regulation (EC) No 1272/2008                                                                                                    |
|-----------|-----------|-------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phenol    | 108-95-2  | EEC No. 203-632-7 | 77.0-81.5 | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Muta. 2 (H341)<br>STOT RE 2 (H373) |
| Water     | 7732-18-5 | 231-791-2         | 18.5-23   | -                                                                                                                                                    |

| Component | Specific concentration limits (SCL's)                                                                | M-Factor | Component notes |
|-----------|------------------------------------------------------------------------------------------------------|----------|-----------------|
| Phenol    | Eye Irrit. 2 (H319) :: 1%<=C<3%<br>Skin Corr. 1B (H314) :: C>=3%<br>Skin Irrit. 2 (H315) :: 1%<=C<3% | -        | -               |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                              |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

### 4.3. Indication of any immediate medical attention and special treatment needed

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## Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

### 6.2. Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

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## 7.2. Conditions for safe storage, including any incompatibilities

Keep refrigerated. Keep container tightly closed. Keep away from heat, sparks and flame. Keep away from oxidizing agents.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1C

**Switzerland - Storage of hazardous substances**

Storage class - SC 6.1

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

The manufacturer recommends a 5 ppm PEL. List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component | European Union                                                                                                        | The United Kingdom                                                                                                    | France                                                                                                                                                                                                                       | Belgium                                                                                                                         | Spain                                                                                                                                                                           |
|-----------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phenol    | TWA: 2 ppm (8h)<br>TWA: 8 mg/m <sup>3</sup> (8h)<br>STEL: 4 ppm (15min)<br>STEL: 16 mg/m <sup>3</sup> (15min)<br>Skin | STEL: 4 ppm 15 min<br>STEL: 16 mg/m <sup>3</sup> 15 min<br>TWA: 2 ppm 8 hr<br>TWA: 7.8 mg/m <sup>3</sup> 8 hr<br>Skin | TWA / VME: 2 ppm (8 heures). restrictive limit<br>TWA / VME: 7.8 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 4 ppm. restrictive limit<br>STEL / VLCT: 15.6 mg/m <sup>3</sup> . restrictive limit<br>Peau | TWA: 2 ppm 8 uren<br>TWA: 8 mg/m <sup>3</sup> 8 uren<br>STEL: 4 ppm 15 minuten<br>STEL: 16 mg/m <sup>3</sup> 15 minuten<br>Huid | STEL / VLA-EC: 4 ppm (15 minutos).<br>STEL / VLA-EC: 16 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 2 ppm (8 horas)<br>TWA / VLA-ED: 8 mg/m <sup>3</sup> (8 horas)<br>Piel |

| Component | Italy                                                                                                                                                                                                | Germany                                                                                                                  | Portugal                                                                                                                          | The Netherlands                         | Finland                                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Phenol    | TWA: 2 ppm 8 ore. Time Weighted Average<br>TWA: 8.0 mg/m <sup>3</sup> 8 ore. Time Weighted Average<br>STEL: 4 ppm 15 minuti. Short-term<br>STEL: 16 mg/m <sup>3</sup> 15 minuti. Short-term<br>Pelle | TWA: 2 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 8 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>Haut | STEL: 4 ppm 15 minutos<br>STEL: 16 mg/m <sup>3</sup> 15 minutos<br>TWA: 2 ppm 8 horas<br>TWA: 8 mg/m <sup>3</sup> 8 horas<br>Pele | huid<br>TWA: 8 mg/m <sup>3</sup> 8 uren | TWA: 2 ppm 8 tunteina<br>TWA: 8 mg/m <sup>3</sup> 8 tunteina<br>STEL: 4 ppm 15 minuutteina<br>STEL: 16 mg/m <sup>3</sup> 15 minuutteina<br>Iho |

| Component | Austria                                                                                                                                               | Denmark                                                                                                                            | Switzerland                                                                                                                                 | Poland                                                                           | Norway                                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phenol    | Haut<br>MAK-KZGW: 4 ppm 15 Minuten<br>MAK-KZGW: 16 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 2 ppm 8 Stunden<br>MAK-TMW: 8 mg/m <sup>3</sup> 8 Stunden | TWA: 1 ppm 8 timer<br>TWA: 4 mg/m <sup>3</sup> 8 timer<br>STEL: 16 mg/m <sup>3</sup> 15 minutter<br>STEL: 4 ppm 15 minutter<br>Hud | Haut/Peau<br>STEL: 5 ppm 15 Minuten<br>STEL: 19 mg/m <sup>3</sup> 15 Minuten<br>TWA: 5 ppm 8 Stunden<br>TWA: 19 mg/m <sup>3</sup> 8 Stunden | STEL: 16 mg/m <sup>3</sup> 15 minutach<br>TWA: 7.8 mg/m <sup>3</sup> 8 godzinach | TWA: 1 ppm 8 timer<br>TWA: 4 mg/m <sup>3</sup> 8 timer<br>STEL: 3 ppm 15 minutter. value from the regulation<br>STEL: 12 mg/m <sup>3</sup> 15 minutter. value from the regulation<br>Hud |

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| Component | Bulgaria                                                                                               | Croatia                                                                                                                                                     | Ireland                                                                                                               | Cyprus                                                                                                                         | Czech Republic                                                                                                |
|-----------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Phenol    | TWA: 2 ppm<br>TWA: 8 mg/m <sup>3</sup><br>STEL : 4 ppm<br>STEL : 16 mg/m <sup>3</sup><br>Skin notation | kože<br>TWA-GVI: 2 ppm 8 satima.<br>TWA-GVI: 8 mg/m <sup>3</sup> 8 satima.<br>STEL-KGVI: 4 ppm 15 minutama.<br>STEL-KGVI: 16 mg/m <sup>3</sup> 15 minutama. | TWA: 2 ppm 8 hr.<br>TWA: 8 mg/m <sup>3</sup> 8 hr.<br>STEL: 4 ppm 15 min<br>STEL: 16 mg/m <sup>3</sup> 15 min<br>Skin | Skin-potential for cutaneous absorption<br>STEL: 16 mg/m <sup>3</sup><br>STEL: 4 ppm<br>TWA: 8 mg/m <sup>3</sup><br>TWA: 2 ppm | TWA: 7.5 mg/m <sup>3</sup> 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 15 mg/m <sup>3</sup> |

| Component | Estonia                                                                                                                                         | Gibraltar                                                                                                                    | Greece                                                                                                                           | Hungary                                                                                                                        | Iceland                                                                                                                                     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Phenol    | Nahk<br>TWA: 2 ppm 8 tundides.<br>TWA: 8 mg/m <sup>3</sup> 8 tundides.<br>STEL: 16 mg/m <sup>3</sup> 15 minutites.<br>STEL: 4 ppm 15 minutites. | Skin notation<br>TWA: 2 ppm 8 hr<br>TWA: 8 mg/m <sup>3</sup> 8 hr<br>STEL: 16 mg/m <sup>3</sup> 15 min<br>STEL: 4 ppm 15 min | skin - potential for cutaneous absorption<br>STEL: 4 ppm<br>STEL: 16 mg/m <sup>3</sup><br>TWA: 2 ppm<br>TWA: 8 mg/m <sup>3</sup> | STEL: 16 mg/m <sup>3</sup> 15 percekben. CK<br>TWA: 8 mg/m <sup>3</sup> 8 órában. AK<br>lehetséges borón keresztül felszívódás | TWA: 1 ppm 8 klukkustundum.<br>TWA: 4 mg/m <sup>3</sup> 8 klukkustundum.<br>Skin notation<br>Ceiling: 2 ppm<br>Ceiling: 8 mg/m <sup>3</sup> |

| Component | Latvia                                                                                                                         | Lithuania                                                                                            | Luxembourg                                                                                                                                                                          | Malta                                                                                                                                                         | Romania                                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Phenol    | skin - potential for cutaneous exposure<br>STEL: 4 ppm<br>STEL: 16 mg/m <sup>3</sup><br>TWA: 2 ppm<br>TWA: 8 mg/m <sup>3</sup> | TWA: 2 ppm IPRD<br>TWA: 8 mg/m <sup>3</sup> IPRD<br>Oda<br>STEL: 4 ppm<br>STEL: 16 mg/m <sup>3</sup> | Possibility of significant uptake through the skin<br>TWA: 2 ppm 8 Stunden<br>TWA: 8 mg/m <sup>3</sup> 8 Stunden<br>STEL: 16 mg/m <sup>3</sup> 15 Minuten<br>STEL: 4 ppm 15 Minuten | possibility of significant uptake through the skin<br>TWA: 2 ppm<br>TWA: 8 mg/m <sup>3</sup><br>STEL: 16 mg/m <sup>3</sup> 15 minuti<br>STEL: 4 ppm 15 minuti | Skin notation<br>TWA: 2 ppm 8 ore<br>TWA: 8 mg/m <sup>3</sup> 8 ore<br>STEL: 4 ppm 15 minute<br>STEL: 16 mg/m <sup>3</sup> 15 minute |

| Component | Russia                                                                       | Slovak Republic                                                                                               | Slovenia                                                                                                                        | Sweden                                                                                                                                                             | Turkey                                                                                                                        |
|-----------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Phenol    | TWA: 0.3 mg/m <sup>3</sup> 0539<br>Skin notation<br>MAC: 1 mg/m <sup>3</sup> | Ceiling: 16 mg/m <sup>3</sup><br>Potential for cutaneous absorption<br>TWA: 2 ppm<br>TWA: 8 mg/m <sup>3</sup> | TWA: 2 ppm 8 urah<br>TWA: 8 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 4 ppm 15 minutah<br>STEL: 16 mg/m <sup>3</sup> 15 minutah | Binding STEL: 4 ppm 15 minuter<br>Binding STEL: 16 mg/m <sup>3</sup> 15 minuter<br>TLV: 1 ppm 8 timmar.<br>NGV<br>TLV: 4 mg/m <sup>3</sup> 8 timmar.<br>NGV<br>Hud | Deri<br>TWA: 2 ppm 8 saat<br>TWA: 8 mg/m <sup>3</sup> 8 saat<br>STEL: 4 ppm 15 dakika<br>STEL: 16 mg/m <sup>3</sup> 15 dakika |

## Biological limit values

List source(s):

| Component | European Union | United Kingdom | France                                               | Spain                                    | Germany                                                              |
|-----------|----------------|----------------|------------------------------------------------------|------------------------------------------|----------------------------------------------------------------------|
| Phenol    |                |                | Total Phenol: 250 mg/g creatinine urine end of shift | : 120 mg/g Creatinine urine end of shift | Phenol (after hydrolysis): 120 mg/g Creatinine urine (end of shift ) |

| Component | Italy | Finland                                         | Denmark | Bulgaria                                                           | Romania                                              |
|-----------|-------|-------------------------------------------------|---------|--------------------------------------------------------------------|------------------------------------------------------|
| Phenol    |       | Total phenol: 1.3 mmol/L urine after the shift. |         | Phenol: 200 µg/L urine at the end of exposure or end of work shift | total Phenol: 120 mg/g Creatinine urine end of shift |

| Component | Gibraltar | Latvia | Slovak Republic                                      | Luxembourg | Turkey |
|-----------|-----------|--------|------------------------------------------------------|------------|--------|
| Phenol    |           |        | Phenol: 200 mg/L urine end of exposure or work shift |            |        |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

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## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                        | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|----------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Phenol<br>108-95-2 ( 77.0-81.5 ) |                              |                                 |                                | DNEL = 1.23mg/kg bw/day           |

| Component                        | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|----------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Phenol<br>108-95-2 ( 77.0-81.5 ) | DNEL = 16mg/m <sup>3</sup>       |                                     |                                    | DNEL = 8mg/m <sup>3</sup>             |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                        | Fresh water       | Fresh water sediment           | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)        |
|----------------------------------|-------------------|--------------------------------|--------------------|------------------------------------|---------------------------|
| Phenol<br>108-95-2 ( 77.0-81.5 ) | PNEC = 0.0077mg/L | PNEC = 0.0915mg/kg sediment dw | PNEC = 0.031mg/L   | PNEC = 2.1mg/L                     | PNEC = 0.136mg/kg soil dw |

| Component                        | Marine water       | Marine water sediment           | Marine water Intermittent | Food chain | Air |
|----------------------------------|--------------------|---------------------------------|---------------------------|------------|-----|
| Phenol<br>108-95-2 ( 77.0-81.5 ) | PNEC = 0.00077mg/L | PNEC = 0.00915mg/kg sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Neoprene       |                                   |                 |             |                       |
| Natural rubber |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

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|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Organic gases and vapours filter Type A Brown conforming to EN14387                                                                  |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.                                                                                                                                           |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              | Light red                |                                          |
| <b>Odor</b>                                    | Strong                   |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>Melting Point/Range</b>                     | 40.5 °C / 104.9 °F       |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flammability (liquid)</b>                   | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | Not applicable           | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>pH</b>                                      | ≈5                       |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | Soluble                  |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Phenol                                         | 1.5                      |                                          |
| <b>Vapor Pressure</b>                          | 1 mmHg @ 40 °C           |                                          |
| <b>Density / Specific Gravity</b>              | 1.06                     |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | 3.2                      | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                          |

### 9.2. Other information

|                         |                              |
|-------------------------|------------------------------|
| <b>Evaporation Rate</b> | < 0.01 (Butyl Acetate = 1.0) |
|-------------------------|------------------------------|

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability



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Stable under normal conditions.

## 10.3. Possibility of hazardous reactions

**Hazardous Polymerization** Hazardous polymerization does not occur.  
**Hazardous Reactions** None under normal processing.

## 10.4. Conditions to avoid

Incompatible products. Excess heat.

## 10.5. Incompatible materials

Strong oxidizing agents.

## 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

(a) acute toxicity;  
Oral Category 3  
Dermal Category 3  
Inhalation Category 3

#### Toxicology data for the components

| Component | LD50 Oral                | LD50 Dermal                 | LC50 Inhalation                          |
|-----------|--------------------------|-----------------------------|------------------------------------------|
| Phenol    | LD50 = 340 mg/kg ( Rat ) | LD50 = 630 mg/kg ( Rabbit ) | LC50 = 316 mg/m <sup>3</sup> ( Rat ) 4 h |
| Water     | -                        | -                           | -                                        |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; Category 2

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | EU | UK | Germany | IARC |
|-----------|----|----|---------|------|
| Phenol    |    |    | Cat. 3B |      |

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

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(i) STOT-repeated exposure; Category 2

Target Organs Kidney, Liver, Skin, Central nervous system (CNS).

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

## 11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component | Freshwater Fish                         | Water Flea                                                                                              | Freshwater Algae                                                                                                                                                                                                   |
|-----------|-----------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phenol    | 4-7 mg/L LC50 96 h<br>32 mg/L LC50 96 h | EC50: 10.2 - 15.5 mg/L, 48h<br>(Daphnia magna)<br>EC50: 4.24 - 10.7 mg/L, 48h<br>Static (Daphnia magna) | EC50: 187 - 279 mg/L, 72h<br>static (Desmodesmus<br>subspicatus)<br>EC50: 0.0188 - 0.1044 mg/L,<br>96h static (Pseudokirchneriella<br>subcapitata)<br>EC50: = 46.42 mg/L, 96h<br>(Pseudokirchneriella subcapitata) |

| Component | Microtox                                                                                                                             | M-Factor |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------|----------|
| Phenol    | EC50 21 - 36 mg/L 30 min<br>EC50 = 23.28 mg/L 5 min<br>EC50 = 25.61 mg/L 15 min<br>EC50 = 28.8 mg/L 5 min<br>EC50 = 31.6 mg/L 15 min |          |

### 12.2. Persistence and degradability

Persistence Soluble in water, Persistence is unlikely, based on information available.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF)           |
|-----------|---------|-----------------------------------------|
| Phenol    | 1.5     | 17.5 dimensionless<br>647 dimensionless |

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

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**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

**Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

14.1. UN number

UN2821

14.2. UN proper shipping name

PHENOL SOLUTION

14.3. Transport hazard class(es)

6.1

14.4. Packing group

II

### ADR

14.1. UN number

UN2821

14.2. UN proper shipping name

PHENOL SOLUTION

14.3. Transport hazard class(es)

6.1

14.4. Packing group

II

### IATA

14.1. UN number

UN2821

14.2. UN proper shipping name

PHENOL SOLUTION

14.3. Transport hazard class(es)

6.1

14.4. Packing group

II

14.5. Environmental hazards

No hazards identified

14.6. Special precautions for user

No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

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## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Phenol    | 108-95-2  | 203-632-7 | -      | -   | X     | X    | X        | X    | X    |
| Water     | 7732-18-5 | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |

| Component | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Phenol    | 108-95-2  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Water     | 7732-18-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

| Component | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Phenol    | 108-95-2  | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Water     | 7732-18-5 | -                                                                   | -                                                                             | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

#### Seveso III Directive (2012/18/EC)

| Component | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Phenol    | 108-95-2  | Not applicable                                                                            | Not applicable                                                                           |
| Water     | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           |

#### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

#### Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

#### National Regulations

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UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** Water endangering class = 2 (self classification)

| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                              |
|-----------|---------------------------------------|------------------------------------------------------|
| Phenol    | WGK2                                  | Class I : 20 mg/m <sup>3</sup> (Massenkonzentration) |

| Component | France - INRS (Tables of occupational diseases)      |
|-----------|------------------------------------------------------|
| Phenol    | Tableaux des maladies professionnelles (TMP) - RG 14 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                        | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Phenol<br>108-95-2 ( 77.0-81.5 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H301 - Toxic if swallowed  
H311 - Toxic in contact with skin  
H331 - Toxic if inhaled  
H314 - Causes severe skin burns and eye damage  
H318 - Causes serious eye damage  
H341 - Suspected of causing genetic defects  
H373 - May cause damage to organs through prolonged or repeated exposure

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

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OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Creation Date** 23-May-2011

**Revision Date** 05-Jun-2024

**Revision Summary** SDS sections updated, 3, 7.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**